

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Touch Display
Manufacturer: CleverProducts
Firmware Version: N/A
Model(s): Clevertouch Plus 42", Clevertouch Plus 55", Clevertouch Plus 65", Clevertouch Plus 70",
Clevertouch Plus 84", Clevertouch Plus 75"

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.2

Version History

Module Version	Date	Notes
1_0_2_0	12/16/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5
- For Serial Over Ethernet support, SerialOverEthernetClass must be used. Example: InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', PortNum, Model='ModelName')

Control Commands, States, and Qualifiers:

Aspect	'16:9'	'4:3'	'Point to point'
AudioMute	'On'	'Off'	
Channels	'TV' 'VGA1' 'HDMI2' 'Android'	'CVBS' 'VGA2' 'HDMI3' 'HDMI (4K*2K)'	'VGA3' 'HDMI1' 'PC'
FunctionsofRemoteControl	'WIN' 'Alt+F4' 'NUM_3' 'NUM_6' 'NUM_9' 'Refresh' 'Menu' 'UP' 'RIGHT' 'Back' 'VOL+' 'PageDown' 'F3' 'F6' 'F9' 'F12' 'YELLOW'	'Space' 'NUM_1' 'NUM_4' 'NUM_7' 'NUM_0' 'Input' 'Delete' 'DOWN' 'ENTER' 'CH+ ' 'VOL- ' 'F1' 'F4' 'F7' 'F10' 'RED' 'BLUE'	'Alt+Tab' 'NUM_2' 'NUM_5' 'NUM_8' 'Display' 'Home' 'Energy' 'LEFT' 'Point' 'CH- ' 'PageUp' 'F2' 'F5' 'F8' 'F11' 'GREEN'
Power	'On'	'Off'	
PowerofPC	'On'	'Off'	
TVchannels	0 to 99 in steps of 1		
Volume	0 to 100 in steps of 1		

Status Available:

AudioMute	'On'	'Off'	
ConnectionStatus	'Connected'	'Disconnected'	
InputStatus	'TV'	'AV'	'VGA 1'
	'VGA 2'	'VGA 3'	'HDMI 1'
	'HDMI 2'	'HDMI 3'	'PC'
	'Android'	'HDMI (4K*2K)'	'WHDI'
Power	'On'	'Off'	
PowerofPC	'On'	'Off'	'Sleep'
	'Hibernate'		
TVchannels	0 to 99 in steps of 1		
Volume	0 to 100 in steps of 1		

Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device:

Serial communication:

Port Type: RS-232

Baud Rate: 9600

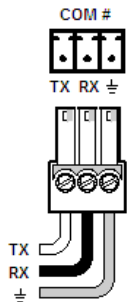
Data Bits: 8

Parity: None

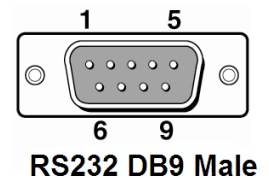
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



General Notes:

Appendix A. Set Commands

Aspect 16:9	\xAA\xBB\xCC\x08\x00\x00\x08\xDD\xEE\xFF
Aspect 4:3	\xAA\xBB\xCC\x08\x01\x00\x09\xDD\xEE\xFF
Aspect Point to point	\xAA\xBB\xCC\x08\x07\x00\x0F\xDD\xEE\xFF
Audio Mute Off	\xAA\xBB\xCC\x03\x01\x01\x05\xDD\xEE\xFF
Audio Mute On	\xAA\xBB\xCC\x03\x01\x00\x04\xDD\xEE\xFF
Channels Android	\xAA\xBB\xCC\x02\x0A\x00\x0C\xDD\xEE\xFF
Channels CVBS	\xAA\xBB\xCC\x02\x02\x00\x04\xDD\xEE\xFF
Channels HDMI (4K*2K)	\xAA\xBB\xCC\x02\x0D\x00\x0F\xDD\xEE\xFF
Channels HDMI1	\xAA\xBB\xCC\x02\x06\x00\x08\xDD\xEE\xFF
Channels HDMI2	\xAA\xBB\xCC\x02\x07\x00\x09\xDD\xEE\xFF
Channels HDMI3	\xAA\xBB\xCC\x02\x05\x00\x07\xDD\xEE\xFF
Channels PC	\xAA\xBB\xCC\x02\x08\x00\x0A\xDD\xEE\xFF
Channels TV	\xAA\xBB\xCC\x02\x01\x00\x03\xDD\xEE\xFF
Channels VGA1	\xAA\xBB\xCC\x02\x03\x00\x05\xDD\xEE\xFF
Channels VGA2	\xAA\xBB\xCC\x02\x04\x00\x06\xDD\xEE\xFF
Channels VGA3	\xAA\xBB\xCC\x02\x0B\x00\x0D\xDD\xEE\xFF
Functions of Remote Control Alt+F4	\xAA\xBB\xCC\x07\x1F\x00&\xDD\xEE\xFF
Functions of Remote Control Alt+Tab	\xAA\xBB\xCC\x07\x1D\x00\$\xDD\xEE\xFF
Functions of Remote Control Back	\xAA\xBB\xCC\x07\x0A\x00\x11\xDD\xEE\xFF
Functions of Remote Control BLUE	\xAA\xBB\xCC\x07_\x00f\xDD\xEE\xFF
Functions of Remote Control CH-	\xAA\xBB\xCC\x07\x09\x00\x10\xDD\xEE\xFF
Functions of Remote Control CH+	\xAA\xBB\xCC\x07\x02\x00\x09\xDD\xEE\xFF
Functions of Remote Control Delete	\xAA\xBB\xCC\x07@\x00G\xDD\xEE\xFF
Functions of Remote Control Display	\xAA\xBB\xCC\x07\x1C\x00#\xDD\xEE\xFF
Functions of Remote Control DOWN	\xAA\xBB\xCC\x07M\x00T\xDD\xEE\xFF
Functions of Remote Control Energy	\xAA\xBB\xCC\x07N\x00U\xDD\xEE\xFF
Functions of Remote Control ENTER	\xAA\xBB\xCC\x07J\x00Q\xDD\xEE\xFF
Functions of Remote Control F1	\xAA\xBB\xCC\x07E\x00L\xDD\xEE\xFF
Functions of Remote Control F10	\xAA\xBB\xCC\x07Y\x00`\xDD\xEE\xFF
Functions of Remote Control F11	\xAA\xBB\xCC\x07W\x00^\xDD\xEE\xFF
Functions of Remote Control F12	\xAA\xBB\xCC\x07\x08\x00\x0F\xDD\xEE\xFF
Functions of Remote Control F2	\xAA\xBB\xCC\x07\x12\x00\x19\xDD\xEE\xFF
Functions of Remote Control F3	\xAA\xBB\xCC\x07Q\x00X\xDD\xEE\xFF
Functions of Remote Control F4	\xAA\xBB\xCC\x07[\x00b\xDD\xEE\xFF
Functions of Remote Control F5	\xAA\xBB\xCC\x07D\x00K\xDD\xEE\xFF
Functions of Remote Control F6	\xAA\xBB\xCC\x07P\x00W\xDD\xEE\xFF
Functions of Remote Control F7	\xAA\xBB\xCC\x07C\x00J\xDD\xEE\xFF
Functions of Remote Control F8	\xAA\xBB\xCC\x07\x1A\x00!\xDD\xEE\xFF
Functions of Remote Control F9	\xAA\xBB\xCC\x07\x04\x00\x0B\xDD\xEE\xFF
Functions of Remote Control GREEN	\xAA\xBB\xCC\x07j\x00d\xDD\xEE\xFF
Functions of Remote Control Home	\xAA\xBB\xCC\x07H\x00O\xDD\xEE\xFF
Functions of Remote Control Input	\xAA\xBB\xCC\x07\x07\x00\x0E\xDD\xEE\xFF
Functions of Remote Control LEFT	\xAA\xBB\xCC\x07I\x00P\xDD\xEE\xFF
Functions of Remote Control Menu	\xAA\xBB\xCC\x07\x0D\x00\x14\xDD\xEE\xFF
Functions of Remote Control NUM_0	\xAA\xBB\xCC\x07\x1B\x00"\xDD\xEE\xFF

Functions of Remote Control NUM_1	\xAA\xBB\xCC\x07\x00\x00\x07\xDD\xEE\xFF
Functions of Remote Control NUM_2	\xAA\xBB\xCC\x07\x10\x00\x17\xDD\xEE\xFF
Functions of Remote Control NUM_3	\xAA\xBB\xCC\x07\x11\x00\x18\xDD\xEE\xFF
Functions of Remote Control NUM_4	\xAA\xBB\xCC\x07\x13\x00\x1A\xDD\xEE\xFF
Functions of Remote Control NUM_5	\xAA\xBB\xCC\x07\x14\x00\x1B\xDD\xEE\xFF
Functions of Remote Control NUM_6	\xAA\xBB\xCC\x07\x15\x00\x1C\xDD\xEE\xFF
Functions of Remote Control NUM_7	\xAA\xBB\xCC\x07\x17\x00\x1E\xDD\xEE\xFF
Functions of Remote Control NUM_8	\xAA\xBB\xCC\x07\x18\x00\x1F\xDD\xEE\xFF
Functions of Remote Control NUM_9	\xAA\xBB\xCC\x07\x19\x00 \xDD\xEE\xFF
Functions of Remote Control PageDown	\xAA\xBB\xCC\x07\x0F\x00\x16\xDD\xEE\xFF
Functions of Remote Control PageUp	\xAA\xBB\xCC\x07B\x00I\xDD\xEE\xFF
Functions of Remote Control Point	\xAA\xBB\xCC\x07\x06\x00\x0D\xDD\xEE\xFF
Functions of Remote Control RED	\xAA\xBB\xCC\x07\x00c\xDD\xEE\xFF
Functions of Remote Control Refresh	\xAA\xBB\xCC\x07L\x00S\xDD\xEE\xFF
Functions of Remote Control RIGHT	\xAA\xBB\xCC\x07K\x00R\xDD\xEE\xFF
Functions of Remote Control Space	\xAA\xBB\xCC\x07F\x00M\xDD\xEE\xFF
Functions of Remote Control UP	\xAA\xBB\xCC\x07G\x00N\xDD\xEE\xFF
Functions of Remote Control VOL-	\xAA\xBB\xCC\x07A\x00H\xDD\xEE\xFF
Functions of Remote Control VOL+	\xAA\xBB\xCC\x07\x03\x00\x0A\xDD\xEE\xFF
Functions of Remote Control WIN	\xAA\xBB\xCC\x07\x0B\x00\x12\xDD\xEE\xFF
Functions of Remote Control YELLOW	\xAA\xBB\xCC\x07^\x00e\xDD\xEE\xFF
Power of PC Off	\xAA\xBB\xCC\x09\x00\x00\x09\xDD\xEE\xFF
Power of PC On	\xAA\xBB\xCC\x09\x01\x00\x0A\xDD\xEE\xFF
Power Off	\xAA\xBB\xCC\x01\x01\x00\x02\xDD\xEE\xFF
Power On	\xAA\xBB\xCC\x01\x00\x00\x01\xDD\xEE\xFF
TV channels 0	\xAA\xBB\xCC\x05\x00\x00\x05\xDD\xEE\xFF
Volume 0	\xAA\xBB\xCC\x03\x00\x00\x03\xDD\xEE\xFF

Appendix B. Query Commands

Audio Mute None	\xAA\xBB\xCC\x03\x03\x00\x06\xDD\xEE\xFF
Input Status None	\xAA\xBB\xCC\x02\x00\x00\x02\xDD\xEE\xFF
Power None	\xAA\xBB\xCC\x01\x02\x00\x03\xDD\xEE\xFF
Power of PC None	\xAA\xBB\xCC\x09\x02\x00\x0B\xDD\xEE\xFF
Volume None	\xAA\xBB\xCC\x03\x02\x00\x05\xDD\xEE\xFF